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SUMMARY

Senior engineer building the control plane for enterprise AI: LLM gateways with per-tenant budgets, policy-as-code (OPA/Rego) enforced fail-closed at runtime, non-human identity for agents, per-identity usage and spend attribution, red-team and eval gates before deploy, and tamper-evident audit trails across humans, agents, and tool calls. 4+ years shipping production platforms end-to-end at startups and in BFSI - TypeScript, Golang, Python, Kubernetes, AWS, Keycloak/OAuth. Currently building agentic, intelligent-automation, and knowledge-graph platforms at Mphasis, and independently built and launched **Praxis**, a live platform to build, prove, and run governed AI agents.

INDEPENDENT PRODUCT

- **Praxis - The Platform for Your AI Workforce** praxis.shreyasmk.fyi
 - *Solo builder · Golang, TypeScript, AgentScope, LiteLLM, Temporal, OPA, Keycloak*
 - **Identity-aware gateway:** Every LLM and tool call passes one chokepoint (GATE) that verifies identity, enforces role-based authorization, runs inline prompt-injection/jailbreak guardrails, and signs an audit event. Each agent is a non-human identity minting short-lived tokens: no standing keys, forged actor headers ignored, suspension revokes token issuance instantly. Joiner/mover/leaver lifecycle, periodic access reviews, and SCIM 2.0 for enterprise IGA tooling.
 - **Attribution & audit:** Token spend and guardrail events attributed to the exact human or agent identity; provider-agnostic model routing with budgets via LiteLLM. Humans, agents, and service calls land on one signed, tamper-evident, object-locked timeline, queryable via an audit API for who did what, under which identity and role, against which model and data.
 - **Proof before production:** WARDEN red-team gate runs adversarial and functional tests tagged to OWASP LLM/Agentic Top 10, MITRE ATT&CK, GDPR, and the EU AI Act, and can block a deploy; Agent Evals grade runs on versioned datasets with optimization runs committable as the new baseline; a Policy Engine compiles natural-language rules to Rego, enforced fail-closed by OPA.
 - **Build & run:** Agents are declared as a versioned spec plus role file and generated onto AgentScope, each running as its own service; tools flagged for confirmation pause for human approve/deny. Workflow Composer runs business processes as durable Temporal graphs (agent, HTTP/MCP, rule-based decision, and human-task nodes), with a human gate on every state-changing external call. Driven by a CLI and an installable builder skill for Claude Code / Cursor.

EXPERIENCE

- **Mphasis** Bangalore, India
 - *Senior Software Engineer, Applied AI* Feb 2026 - Present
 - **ATOM - Enterprise Agentic Platform:** *TypeScript, Golang, FastAPI, React, Kubernetes, Kafka (Redpanda), OPA, PostgreSQL, Redis*
Architected and shipped a multi-tenant platform for BFSI clients - TypeScript/Golang/FastAPI services, a web Studio UI, and a policy gateway (auth, OPA, rate-limiting, immutable audit log) - now running 500+ agents at 1M+ requests/day. Built AEGIS, its security layer, which red-teams agents (prompt injection, jailbreak, goal-hijacking) and enforces PCI-DSS/GDPR/DORA guardrails.
 - **OathKeeper - Intelligent RPA for BFSI Operations (in progress):** *RPA across Java, C#, Web, Mail & Mainframe; PII redaction; audit; foundation-model training*
Today an SOP can be automated with RPA, but the intelligence is lost: the factors that made a BPS operator take a decision never reach the bot. OathKeeper keeps that oath. It captures operators across Java, C#, web, mail, and mainframe applications, with redaction at capture and every step audited, and moves each process through record, replay, shadow, and takeover. The recorded behaviour, screenshots, and sessions then train a foundation model to decide the next action, cutting human-in-the-loop to the calls that need a person.
 - **Ontologic - AI-Co-Designed Knowledge Graph for Enterprise Intelligence:** *RDF/OWL, Turtle, SHACL, Apache Jena, ArcadeDB, LLMs*
Designing a system where domain experts and an AI co-designer build a domain ontology together: the AI proposes classes and relations, a named human confirms each definition, and the approved model is emitted as a validated TTL file and deployed onto a data-intelligence platform with ArcadeDB as the graph store and Apache Jena as the reasoning layer. Modeled bi-temporal provenance (valid time vs. belief time), first-class relations, a single validated write path, and append-only corrections, so the system can answer what it believed at any past moment, what is missing, and where sources disagree.
- **IHX - A Perfios Company** Bangalore, India
 - *Senior Software Engineer* Jul 2024 - Feb 2026

- **Persistence Engine:** *TypeScript, TypeORM, PostgreSQL, Redis*
Architected a single-layer persistence solution that cut average query time from 1-2s to 10-15ms across 10 tables.
- **Status Notification Service:** *TypeScript, AWS SQS/SNS, MongoDB*
Built a high-throughput system processing 10M+ messages/day to track state across AI/ML entities, surfacing unified real-time status to clients.
- **Workflow Orchestration Engine:** *Netflix Conductor, Zookeeper, Redis Sentinel, TypeScript*
Architected orchestration for document pipelines (extraction, OCR, classification, QC) with parallel processing, cutting processing time 40%.

IHX - A Perfios Company

Bangalore, India

Aug 2022 - Jul 2024

• Software Engineer

- **Document Intelligence Platform:** *TypeScript, FastAPI, AWS EKS, MongoDB, S3*
Built the document-processing platform for India's largest health benefits administrator, lifting first-instance complete claim submissions from 23% to 77% within 2 weeks; now classifies 4M+ medical images a month.
- **Platform Infrastructure & Identity (End-to-End Ownership):** *Kubernetes, Helm, NGINX, AWS, Grafana, Loki, Prometheus, Keycloak, OAuth 2.0, JWT*
As an early engineer, owned infrastructure and access control end-to-end: Kubernetes/Helm scaling and hardening for AI-model serving, the full observability stack, and a Keycloak MFA/SSO solution with custom plugins that cut login time 50%.

TECHNICAL SKILLS

- **AI Governance & Security:** LLM gateway design, OPA / Rego policy-as-code, agent identity & short-lived service tokens, SCIM 2.0, access reviews, red-teaming (OWASP LLM/Agentic Top 10, MITRE ATT&CK), PII-aware routing, budget governance, immutable audit logging
- **Agents & LLMs:** Multi-agent orchestration, agent evals & grading, human-in-the-loop workflows, intelligent RPA (record, replay, shadow, takeover), RAG, LiteLLM, LangGraph, LangChain, AgentScope
- **Knowledge Graphs:** RDF / OWL ontologies, Turtle (TTL), SHACL, SPARQL, Apache Jena, ArcadeDB, bi-temporal data modeling
- **Languages & Backend:** TypeScript, Golang, Python, SQL; Node.js, FastAPI, Django, Express, REST, WebSockets, Kafka (Redpanda), Temporal, event-driven architecture
- **Data & Infra:** PostgreSQL, MongoDB, Redis; AWS (EKS, SQS, SNS, S3, API Gateway, Cognito), Docker, Kubernetes, Helm, GitHub Actions
- **Identity & Observability:** Keycloak, OAuth 2.0, JWT, MFA, SSO; Grafana, Loki, Tempo, Prometheus

OTHER PROJECTS

- **Collaborative Compiler (JavaScript, WebSockets):** Real-time multi-user code editor with autocomplete, built by hand before AI coding assistants existed. github.com/shreyasY2k/Collaborative-Compiler
- **Contra (Three.js):** Complete 3D run-and-gun shooter built in 2 hours with AI tooling - enemy AI, vehicle physics, particle systems. shreyasy2k.github.io/contra

TECHNICAL WRITING, EDUCATION & CERTIFICATIONS

- **AI Governance & Knowledge Graphs:** *Nobody Left to Ask* (2026) - why automated decisions make provenance the only witness left, and three questions vector/RAG stacks structurally cannot answer. *Your Database Has One Clock. It Needs Two.* (2026) - bi-temporal modeling for security and audit data.
- **Identity:** Multi-part series on Keycloak MFA/SSO (custom plugins, SMS/Email/TOTP/Telegram, centralized user management).
- **Education:** B.E., Information Science, Sapthagiri College of Engineering, Bangalore (2018-2022); CGPA 8.5
- **Certifications:** Architecting with Google Compute Engine (Coursera); The Modern JavaScript Bootcamp (Udemy)